

13 Dec 2025

CS F111 – Computer Programming – End-semester (Open Book, Part-B)
Weightage: 35% Duration: 3 hours

Max. marks: 29

Student Name: _____

ID No.: _____

Room no. _____

Student Sign.: _____

Invigilator Sign.: _____

Marks obtained: _____

/29

Q.1	Q.2	Q.3	Re-check request (if any)
_____	_____	_____	_____

[Instructions: There are total 3 questions. Assume that `#include<stdio.h>` is included at the top of each program. Please WRITE YOUR ANSWERS CLEARLY IN THE SPACE PROVIDED. Answers written at any other place will not be evaluated. Use the rough sheet provided for rough work. There is no negative marking in any question]

[Please note that the outputs are case-sensitive]

[7x2 = 14]

Q.1. Fill the blanks in the following program.

Original Array (Unchanged)				
0	7	5	3	1
10	4	8	2	6
15	12	14	11	13
New Array (Rows Sorted)				
1	3	5	7	9
2	4	6	8	10
Output:	11	12	13	14
				15

$$A = P \left(1 + \frac{r}{n} \right)^n$$

#include <stdlib.h>

```
void bubbleSort(int *arr, int size) {
    int temp, i, j;
    for (i = 0; i < size - 1; i++) {
        for (j = 0; j < size - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }
}
```

```
void createAndSortRows(int rows, int cols, int source[rows][cols], int ***dest) {
    // Allocate memory for the array of row pointers
    *dest = (int **) ?????????????;
    if (*dest == NULL) {
        printf("Memory allocation failed for row pointers.\n");
        return;
    }
    for (int i = 0; i < rows; i++) {
        (*dest)[i] = (int *) ?????????????;
        if ((*dest)[i] == NULL) {
            printf("Memory allocation failed for row %d.\n", i);
            // Cleanup previously allocated memory on failure
            for (int k = 0; k < i; k++)
                ?????????????;
            *dest = NULL;
            return;
        }
        for (int j = 0; j < cols; j++) {
            ?????????????;
        }
        bubbleSort((*dest)[i], cols);
    }
}
```

a). _____ NA

b). _____ NA

c). _____ NA

d). _____ NA

e). _____ NA

```
// Function to print a 2D array
void print2DArray(int rows, int cols, int **arr) {
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < cols; j++) {
            printf("%d\t", arr[i][j]);
        }
    }
}
```

```

    printf("\n");
}
}

// Function to free the memory of a dynamic 2D array
void free2DArray(int rows, int **arr) {
    if (arr == NULL) return;
    // First, free each individual row
    for (int i = 0; i < rows; i++) {
        free(????????);
    }
    // Finally, free the array of row pointers
    free(????????);
}

int main() {
    int original_array[3][5] = {
        {9, 7, 5, 3, 1},
        {10, 4, 8, 2, 6},
        {15, 12, 14, 11, 13}
    };
    int rows = 3;
    int cols = 5;

    // This will be our pointer to the new dynamic 2D array
    int **sorted_array = NULL;

    // We pass the address of our pointer (&sorted_array) so the function can change it.
    createAndSortRows(rows, cols, original_array, &sorted_array);

    printf("--- Original Array (Unchanged) ---\n");

    for (int i = 0; i < rows; i++) { // print original array
        for (int j = 0; j < cols; j++) {
            printf("%d\t", original_array[i][j]);
        }
        printf("\n");
    }

    if (sorted_array != NULL) {
        printf("\n--- New Array (Rows Sorted) ---\n");
        print2DArray(rows, cols, sorted_array);

        // CRITICAL: Free all dynamically allocated memory to prevent leaks
        free2DArray(rows, sorted_array);
    }
    return 0;
}

```

f). _____ NA

g). _____ NA

[6x2 = 12]

Q.2. What is the output of the program?

```

#include <string.h>

void func1(char *start, char *end) {
    while (start < end) {
        char temp = *start;
        *start = *end;
        *end = temp;
        start++;
        end--;
    }
}

void func2(char str[]) {
    char *wordStart = str;
    for (int i = 0; ; i++) {
        if (str[i] == ' ' || str[i] == '\0') {
            func1(wordStart, &str[i - 1]);
            wordStart = &str[i + 1];
        }
        if (str[i] == '\0') break;
    }
}

void func3(char str[]) {
    int len = strlen(str);
    func1(str, str + len - 1);
    func2(str);
}

```

```

int main() {
    char sentence[200] = "Hello how are you doing";

    for (int i = 0; sentence[i] != ' ' && sentence[i] != '\0'; i++)
        putchar(sentence[i]);
    printf("\n");

    int len = strlen(sentence);
    int start = len - 1;
    while (start >= 0 && sentence[start] != ' ')
        start--;
    for (int i = start + 1; i < len; i++)
        putchar(sentence[i]);
    printf("\n");

    start = 0;
    int count = 0;
    for (int i = 0; i <= len; i++) {
        if (sentence[i] == ' ' || sentence[i] == '\0') {
            count++;
            if (count == 2) {
                for (int j = start; j < i; j++)
                    putchar(sentence[j]);
                break;
            }
            start = i + 1;
        }
    }
    printf("\n");

    char temp[200];
    strcpy(temp, sentence);
    func2(temp);
    printf("%s\n", temp);

    char sentenceReversed[200];
    strcpy(sentenceReversed, sentence);
    func3(sentenceReversed);
    printf("%s\n", sentenceReversed);

    start = 0;
    count = 0;
    for (int i = 0; i <= len; i++) {
        if (sentence[i] == ' ' || sentence[i] == '\0') {
            count++;
            if (count == 3) {
                for (int j = start; j < i; j++)
                    putchar(sentence[j]);
                break;
            }
            start = i + 1;
        }
    }
    printf("\n");

    return 0;
}

```

a) hello ✓

b) doing ✓

c) how ✓

d) _____

e) _____

f) _____

NA

(6)

Q.3. What is the output of the program?

[1+2=3]

```

#define max(x) (20)
#define SIZE (10*max(5))

```

```

int main(void)
{
    int k;
    for(k=0; k<max(10); k++);
    printf("%d\n", k);

    for(k=0; k<SIZE; k++);
    printf("%d\n", k);
    return 0;
}

```

a) 20

b) 200

END OF QUESTION PAPER

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